Yerra Akhilesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.85
LAKSHYA SCHOOL	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2024	89.80%
ADITYA (EM) HIGH SCHOOL, GANGARAJU NAGAR	Class X (Board of Secondary Education, ANDHRA PRADESH, INDIA)	2022	95.50%

SKILLS

- **Languages :** C | mySQL | Verilog | Python
- **Essential Technical Skills :** Git | Linux | Latex
- **Tools used :** MATLAB |LTspice |Altium Designer |VS code |Intel Quartus Prime

RELEVANT COURSES

- **Core Courses :** Digital Systems | Analog electronics | Signal Processing | Control Systems | Power Systems | CNA | SDF
- **Lab Courses:** Electrical Circuits Lab | Digital Systems Lab | DSP Lab | Electronic Devices and Circuits Lab | PCB Design
- **Math Courses :** Differential equations and Transform Techniques | Machine Learning | Probability & random processes | Matrix theory
- **Computer Science Courses :** Introduction to programming | Data Structures & Applications | DBMS

PROJECTS

STM32-Based PCB for Diode I-V Characterization

- Designed a custom PCB around the **STM32F303C8T6** featuring DAC buffering, analog signal conditioning using MCP6002 op-amps, regulated power supply, USB-UART interface, and SWD programming support for diode characterization.
- Performed circuit design, PCB layout, component selection, fabrication file generation (Gerber/BOM), and hardware validation to characterize diode electrical behavior under different bias conditions.

UART Transmitter

- Designed and implemented an 8-bit UART Transmitter in Verilog HDL using a **4-state Finite State Machine (FSM)** consisting of IDLE, START, DATA, and STOP states for asynchronous serial communication.
- Implemented a **baud-rate generator**, **shift register**, and **bit counter**, and verified the design using Icarus Verilog and GTKWave, confirming correct serial data transmission and tx_done signal generation.

FPGA-Based FIR Filter & 8-Point FFT Implementation | Verilog , Intel Quartus Prime, MATLAB

- Designed 100-tap FIR filters in Verilog using three architectures (direct form, optimized, and genvar-based) across both non-pipelined and pipelined datapaths, generating filter coefficients in MATLAB and validating outputs against

MATLAB reference results.
- Implemented a pipelined Radix-2 8-point FFT with streaming complex inputs in Verilog HDL and verified functionality using **Intel FFT IP** and MATLAB, including **throughput, Fmax, and resource utilization** analysis.
- Compiled the Verilog designs in Intel Quartus Prime, configured FPGA I/O pin assignments using the Pin Planner, and programmed the Intel DE10-Lite FPGA via USB-Blaster to verify hardware functionality.

Traffic Signal Controller | Verilog

- Designed and implemented a **Finite State Machine (FSM)-based traffic signal controller** in Verilog to regulate traffic flow using timed state transitions.
- Simulated and verified the design for **normal traffic operation, pedestrian crossing requests, and emergency reset conditions** using a Verilog testbench.

8-Bit Breadboard Computer

- Built an 8-bit breadboard computer using **74LS-series TTL logic ICs**, implementing the **clock, registers, ALU, RAM, program counter, instruction register** and **EEPROM**-based microcode control logic, based on the **SAP-1 architecture**.
- Performed hardware debugging and verified **instruction execution, data transfer, and arithmetic operations** by loading **machine-level programs** into memory and validating the expected outputs.

Audio Amplifier (Preamp + Class-AB Power Stage)

- Designed and implemented a two-stage audio amplifier -- a low-noise preamplifier to boost weak microphone-level signals, followed by a Class-AB push-pull power stage to drive an 8 speaker.
- Applied proper transistor biasing and diode-based compensation to minimize crossover distortion, and characterized the amplifier by measuring **THD, PSRR, Gain, Bandwidth and Output Swing** using a function generator and oscilloscope.

POSITIONS OF RESPONSIBILITY

Events & PR Coordinator (SPIC MACAY): Led publicity and PR for 5-6 cultural events, including classical music and dance performances and workshops, with major events attracting 500+ attendees; coordinated with artists and venue logistics to ensure smooth execution.

EXTRACURRICULAR

Hockey: Active NSO hockey player; represented the EE branch in Diesta Hockey, competed for the hostel team in Milan, and was a member of the institute hockey team in the Institute Hockey League.